Curriculum Vitae

Emma Lehmberg, MSc
Ecology & Evolutionary Biology Interdisciplinary Program
Department of Biology
Texas A&M University
205 Butler Hall
College Station, Texas, USA
77840 eslehmberg@tamu.edu

RESEARCH INTERESTS

My dissertation focuses on speciation and adaptation, with special interest in how novel traits and modes of reproduction may or may not facilitate speciation in particular clades of the tree of life. To answer these questions, I use comparative phylogenetic methods to examine interactions between particular phenotypes and their relationship to hybridization. My dissertation work is building a phylogeny using full genomes to examine whether the rapid evolution of placentation has limited historical hybridization and accelerated reproductive isolation in live-bearing fishes (Poeciliidae).

EDUCATION

Note: Mentors have played a significant role in my development as a researcher and member of the scientific community but are not a part of a formal academic assessment body (e.g. committee or advisor)

Ph.D., Ecology & Evolutionary Biology
Thesis topic: Hybridization, Speciation, and Sexual Conflict in live-bearing fishes (Poeciliidae)
Advisors: Gil G. Rosenthal & Heath Blackmon
Mentors: Courtney Fitzpatrick, Jessica Light, Manfred Schartl (Texas State University)
Texas A&M University, College Station, Texas
Expected Graduation Date: August 2023

Master of Science, Ecology & Evolutionary Biology, 2015
Thesis topic: Biogeography of the Weakly Electric Knifefish *Gymnotus carapo* (Teleostei: Gymnotiformes) in the Guianas of South America
Advisor: Nathan R. Lovejoy
University of Toronto, Ontario, Canada

Bachelor of Arts, Honors in English with Major in Biology, 2012
Mentor: Noel Alfonso (Canadian Museum of Nature)
University of Ottawa, Ontario, Canada

RESEARCH EXPERIENCE

2023-2024 *Research Associate*, Department of Biology, University of Padova, Padua, Italy

2022 *Visiting student*, Poeciliid Working Group, Department of Biology, University of Padova, Padua, Italy

2021 *Visiting student*, Buckup Lab, Museu Nacional, Rio de Janeiro, Rio de Janeiro, Brazil

2019	*Visiting student*, Schartl Lab, Julius-Maximilians-Universität Würzburg, Würzburg, Bavaria, Germany
2015-2018	*Curator*, Claude E. Garton Herbarium, Lakehead University, Thunder Bay, Ontario, Canada
2015	*Environmental Consultant*, Basket Lake Camp, Dryden, Ontario, Canada
2012-2014	*Collections and Education Contractor*, Canadian Museum of Nature (CMN), Ottawa, Ontario, Canada
2011-2013	Environmental Monitoring Technician, CMN, Ottawa, Ontario, Canada

TEACHING EXPERIENCE

2020-2021	*Teaching Assistant*, Freshman Biology Honors Labs (BIO112H), Texas A&M University, College Station, TX, USA
2019	*Teaching Assistant*, Introduction to Biology for non-Majors (BIO107), Texas A&M University, College Station, TX, USA
2018	*Teaching Assistant*, Introduction to Biology (BIO111), Texas A&M University, College Station, TX, USA
2015-2018	*Undergraduate Biology Lab Co-ordinator & Technician*, Aboriginal Ethnobotany (BIOL3012), Biology of Fishes (BIOL4212), Animal Biology (BIOL1110), Ornithology (BIOL4231), Aquatic Ecology and Experimental Limnology (BIOL4152), Lakehead University, Thunder Bay, Canada
2013-2015	*Teaching Assistant*, Life on Earth: Unifying Principles (BIOA01), Life on Earth II: Form, Function and Integration (BIOA02), Introduction to Ecology and Evolutionary Biology (BIOB52), Evolutionary Genetics (BIOC16), Special Topics in Biodiversity and Systematics (BIOD52) University of Toronto Scarborough, Toronto, Canada

SCHOLARSHIPS

2020-2023	National Science and Engineering Research Council PGS-D ($63 000 CAN)
2018	Applied Biodiversity Science Diversity Scholarship, Texas A&M University ($4 000 USD)
2014	Frederick P. Ide Scholarship, University of Toronto
2014	Arts and Sciences Fellowship, University of Toronto
2013	Edna Margaret Roberston Scholarship, University of Toronto
2013	E.M. Robertson Scholarship, University of Toronto
2007	Merit Scholarship, University of Ottawa

GRANTS

2021	LinnéSys Systematics Research Fund (1000 £)
2021	Society for Systematic Biology Graduate Student Award ($3 000 USD)

AWARDS

2021	James Fletcher Award for Best Paper, Canadian Field Naturalist
2020	Texas A&M University Ecology and Evolutionary Biology Student Service Award ($200 USD)

PEER-REVIEWED PUBLICATIONS

Toskovich, K., **Lehmberg, E.S.,** and Rennie, M.D. Avian migratory phenology of Northwestern Ontario (Canada) in the face of global climate change. In prep.

Elbassiouny, A.A., Waddell, J.C., Van Nynatten, A., Kolmann, M.A., **Lehmberg, E.S.**, Schott, R.K., Buck, L.T., Grampton, W.G.R., Chang, B.S.W., and Lovejoy, N.R. 'Field transcriptomics' reveals dynamic circadian metabolix reprogramming in electric fishes. Journal of Experimental Biology. *In review*.

Hancock, Z.B., **Lehmberg, E.S.,** and Blackmon, H.B. Phylogenetics in Space: How Continuous Spatial Structure Impacts Tree Inference. Molecular Phylogenetics and Evolution. 2022. DOI: https://doi.org/10.1016/j.ympev.2022.107505

Lehmberg, E.S., McKee, G., and Rennie, M.D. Hiding in plain sight: Using amateur naturalist records and herbarium vouchers to examine climate change effects in the early-blooming vascular plants of Northwestern Ontario, Canada. Canadian Field Naturalist. 2021. DOI: https://doi.org/10.22621/cfn.v135i4.2567

Hancock, Z.B., **Lehmberg, E.S.**, and Bradburd, G.S. Neo-Darwinism still haunts evolutionary theory: A modern perspective on Charlesworth, Lande, and Slatkin (1982). Evolution. 2021. DOI: https://doi.org/10.1111/evo.14268

Lehmberg, E.S., Elbassiouny, A.A., Bloom, D.D., López-Fernández, H., Crampton, W.G.R., and Lovejoy, N.R. Fish biogeography in the "Lost World" of the Guiana Shield: Phylogeography of the weakly electric knifefish Gymnotus carapo (Teleostei: Gymnotidae). Journal of Biogeography. 2018. DOI: https://doi.org/10.1111/jbi.13177

Elbassiouny, A.E., Schott, R., Waddell, J.C., Kolmann, M.A., **Lehmberg, E.S.**, Van Nynatten, A.D., Crampton, W.G.R., Chang, B., and Lovejoy, N.R. Mitochondrial genomes of the South American electric knifefishes (Order Gymnotiformes). Mitochondrial DNA Part B. Vol. 1. 2016. DOI: http://dx.doi.org/10.1080/23802359.2016.1174090

PRESS

22 June 2022 Lakehead University team outlines impact of climate change: https://www.netnewsledger.com/2022/06/22/lakehead-university-team-outlines-impacts-of-climate-change/

NON-PEERED REVIEWED PUBLICATIONS

Lehmberg, E.S. "Choosing our words carefully: challenges in communicating science." *American Society of Naturalists Graduate Council*. January 2020. Blog.

Lehmberg, E.S. "Fifty Years of the Claude Garton Herbarium." *Lakehead University*. Winter 2018. Print.

Lehmberg, E.S. "The Woods were Lovely Dark and Deep." *Canadian Museum of Nature*. September 24, 2012. Blog.

Lehmberg, E.S. and Robillard, C. "Research and collections: How about Stewardship?" *Canadian Museum of Nature*. May 25, 2012. Blog.

Lehmberg, E.S. "Wild at the Core, Human at the Edge." *Canadian Geographic*. June 2010. 36. Print.

Lehmberg, E.S. "An Earth-Shaking Industry?" *Canadian Geographic*. April 2010. 24. Print.

Lehmberg, E.S. "All the World's a Laboratory." *Canadian Geographic*. January/February 2010. 26. Print.
Fick, S. and **Lehmberg, E.S.** "A Crowded Neighbourhood." *Canadian Geographic*. December 2009. 31. Print

---

ORAL PRESENTATIONS

Lehmberg, E.S., Souto Santos, I.C.A, Buckup, P.A., Reznick, D., Blackmon, H. and Rosenthal, G. Hybridization and Sexual Conflict: A Test of Macroevolutionary Theory in the Poeciliinae. 9th Conference of Poeciliid Biologists, Wageningen, Netherlands. October 2022.

Lehmberg, E.S. Phylogenomics of the live-bearing Poeciliinae. EEB Research Blitz. EEB Interdisciplinary Program, Texas A&M University, Texas, USA. February 2022.

Lehmberg, E.S., Travis, J., Reznick, D., and Rosenthal, G. Sexual Conflict & Hybridization: A Phylogenomic Test of Macroevolutionary Predictions. American Society of Naturalists Conference at Asilomar (online). January 2021.

Lehmberg, E.S. Hybridization and Placentation in the live-bearing Poeciliinae. EEB Research Blitz. EEB Interdisciplinary Program, Texas A&M University, Texas, USA. February 2019.

Lehmberg, E.S. Biogeography of the Banded Knifefish on the Guiana Shield. Third Annual Interdisciplinary Research and Discovery Symposium, University of Toronto Scarborough, July 2014

SEMINARS

Lehmberg, E.S. Hybridization and Phylogenomics in the Poeciliiade. Department of Biology, University of Ferrara, Padua, Italy, December 2022.

Lehmberg, E.S. International Research and Researchers in the Natural Sciences, around the world and at UniPD. Department of Biology, University of Padova, Padua, Italy. November 2022.

Lehmberg, E.S. Measuring Hybridization and Introgression in the Eukaryotes: A Question of Methods. Department of Biology, University of Padova, Padua, Italy. November 2022.

Lehmberg, E.S. Hybridization and Sexual Conflict in the Poeciliidae. Poeciliid Working Group, Department of Biology, University of Padova, Padua, Italy. October 2022.

Lehmberg, E.S. Phylogenetics of the live-bearing Poeciliinae. EEB Research Blitz. EEB Interdisciplinary Program, Texas A&M University, Texas, USA. February 2021.

Lehmberg, E.S. Hybridization and Placentation in the live-bearing Poeciliinae. EEB Research Blitz. EEB Interdisciplinary Program, Texas A&M University, Texas, USA. February 2019.

Lehmberg, E.S. Herbarium Research Practices. Annual Biology Research Symposium. Department of Biology, Lakehead University, Thunder Bay. February 2018.

Lehmberg, E.S. The Claude Garton Herbarium. Department of Biology, Lakehead University, Thunder Bay. January 2018.

Lehmberg, E.S. and Malek, L. The Claude Garton Herbarium: Past, Present, & Future. Thunder Bay Field Naturalists seminar, Thunder Bay Field Naturalists, Thunder Bay. September 2017.

Lehmberg, E.S. Natural History of the Thunder Bay Region. Lakehead University guest lecture for Place-Based Education (graduate course), April 2016.

Lehmberg, E.S. Biogeography of the Banded Knifefish on the Guiana Shield. University of Toronto EEB

Seminar, March 2015.

Lehmberg, E.S. Environmental Monitoring Programme Year-End Presentation. Canadian Museum of Nature Seminar. 2011-2013.

POSTER PRESENTATIONS

Lehmberg, E.S., Souto Santos, I.C.A, Buckup, P.A., Reznick, D., and Rosenthal, G. Sexual Conflict & Hybridization: A Phylogenomic Test of Macroevolutionary Predictions. Evolution, Cleveland, Ohio. June 24-28, 2022.

Lehmberg, E.S., Travis, J., Reznick, D., and Rosenthal, G. Battle of the Sexes: How viviparity-driven conflict may preclude hybridization in the live-bearing Poeciliinae. 8th Conference of Poeciliid Biologists, Mexico City, Mexico. September 2019.

Lehmberg, E.S. and Lovejoy, N.R. Biogeography of the Banded Knifefish on the Guiana Shield. Annual Joint Meeting of Ichthyologists and Herpetologists, Chattanooga, TN. July 2014.

PUBLIC PRESENTATIONS

Hamilton, N. and **Lehmberg, E.S.** (co-presenters). Evolution & Ecology of Vertebrates. Collegiate High School, Bryan, Texas, USA. February 2020.

Hancock, Z.B. and **Lehmberg, E.S.** (co-presenters). Do You Know Your Ancestor? Population Genetics & Coalescent Theory. EEB Darwin Day, EEB Interdisciplinary Program, Texas A&M University, College Station, Texas, USA. February 2020.

Ruckman, S. and **Lehmberg, E.S.** (co-presenters) Sexual Selection and Animal Behaviour. EEB Darwin Day, EEB Interdisciplinary Program, Texas A&M University, Texas, USA. February 2019.

Rosenthal, G.G.R., Schumer, M., Powell, D., Jofre Rodrigues, G., Fascinetto Zago, P., Garcia, M., and **Lehmberg, E.S.** (co-presenters). Educational hike. Centro de Investigaciones Científicas de las Huastecas "Aguazarca" Uniendo Ciencia y Servicio (CICHAZ), Calnali, Hidalgo, Mexico. June 2018.

Lehmberg. E.S. Boreal Phenology: public walking tours of Lakehead Region Conservation Authority (Mills Block). Claude Garton Herbarium & Lakehead University, Thunder Bay, Ontario, Canada. March – June 2018.

Lehmberg. E.S. The Herbarium: Where Plants Go to Die & Then Live Forever. Native Access Program, Lakehead University, Thunder Bay. September 2016, 2017.

Lehmberg. E.S. Animal Communication: Weakly Electric Knifefish. Native Child and Family Services of Toronto, University of Toronto Scarborough. July 2015.

Lehmberg. E.S. Animal Communication: Weakly Electric Knifefish. Bring Your Kids to Work (Outreach Day), University of Toronto Scarborough. April 2015.

Lehmberg. E.S., Shymanski, A., MacGregor, L., and Robillard, C. Work in the Woods: Educational tours of CMN trails. Environmental Monitoring Program, Canadian Museum of Nature. Summers 2011,

2012, 2013.

Lehmberg. E.S. Identification and collection of freshwater invertebrates. Environmental Monitoring Program, Canadian Museum of Nature. September 2012

MENTORING

I believe that mentorship is the act of taking on enthusiastic undergraduate students, actively listening to their academic and career goals, and designing projects that help both of us achieve our research aims. I strive to have an open dialogue with students, seeking their input when appropriate. It is especially important to me to create a scientific space in which all people, regardless of race, gender, class, or disability are welcome and can explore questions interesting to them.

Rosenthal Lab

**Students I continue to mentor post-graduation*

Alix Garcia, honors student (2020-2022). Collaborating on a simulation project to test introgression detection using a variety of existing phylogenetic softwares.

*Paola Canales-Lozano**, student worker (2018-2021). Lab collections organization and database management, Rosenthal Lab website management and updates.

Annais Bonilla-Johnson,* honors student (2019-2020). Guidance on designing an automated feeding set up for ichthyological behavioral experiments.

Maria Prado, honors student (2019-2020). Reorganizing and cataloguing of wet fishes collections, database management.

Sara Johnson, honors student (2019-2020). Tissue sampling of preserved fishes, DNA extractions.

Claude Garton Herbarium

Yuen Ting (Lilian) Lam, student worker (2016-2018). Identification of plants, updating voucher database, researching historical phenology using local records.

Jordan Lukacs, student worker (2017-2018). Prototyping and testing of digitization set up.

Hanna Dorval, honors student (2016-2017). Use of vouchers and historical records for honors thesis on local rarity and specialization of orchids. Second reader on her honors thesis.

MEMBERSHIPS

Canadian Society for Ecology & Evolution

Society of Systematic Biologists

American Society of Naturalists

Linnean Society

SERVICE

2019-2020, 2023	*Volunteer*, Texas A&M University Darwin Day, Ecology & Evolutionary Biology Interdisciplinary Program
2022-2023	*Mentor*, EEB Mentor Match
2021-2022	*Chair and graduate representative to the Executive Committee*, American Society of Naturalists Graduate Council
2021-2022	*Graduate member*, American Society of Naturalists Regional Workshop Committee
2021-2022	*Co-organizer*, Bilingual Mentoring Project (with Dr. Henry Arenas-Castro), Evolution, online and Cleveland, Ohio
2019-2022	*Member*, American Society of Naturalists Graduate Council
2022	*Volunteer*, Ecological Integration Symposium, Texas A&M University, College Station, Texas
2021	*Co-Moderator*, Decolonizing Evolution panel (with Dr. Henry Arenas-Castro), Evolution (online)
2019-2021	*General Officer*, Ecology & Evolutionary Biology Interdisciplinary Student Organization, Texas A&M University, College Station, Texas
2020	*Student representative*, Graduate Recruitment Committee, Ecology & Evolutionary Biology Interdisciplinary Program, Texas A&M University, College Station, Texas
2020	*Graduate Co-Organizer*, Introduction to Bioinformatics workshop (with Simon Tye at University of Arkansas), Evolution Community Resources for Early-Career Researchers online conference
2020	*Graduate Co-organizer*, Evolution Community Resources for Early-Career Researchers online conference (replacement for Evolution Joint meeting of ASN, SSE, & SSB)
2020	*Co-ordinator of invited speakers*, Texas A&M University EEB discussion and reading group (online), Texas A&M University, College Station, Texas
2020	*Co-ordinator of Sexual Conflict reading group*, Texas A&M University, UC Riverside, & Columbia Universities, online
2019-2020	*Judge*, Texas State Science and Engineering Fair, College Station, Texas, United States
2015-2018	*DJ*, CILU 102.7fm, Lakehead University, Thunder Bay, Ontario, Canada
2012	*Sea turtle nesting survey volunteer*, Canadian Organization for Tropical Education and Research, San Francisco de Tortuguero, Limón, Pococí, Costa Rica
2010-2012	*Volunteer*, Malacology, Canadian Museum of Nature, Ottawa, Ontario, Canada

REVIEWER

2020	*eLife*
2018	*Copeia*

PROFESSIONAL DEVELOPMENT

2023	*Data Cleaning with OpenRefine*, Data Carpentry, Texas A&M University Institute of Data Science, College Station, Texas, US
2023	*Data Organziation*, Data Carpentry, Texas A&M University Institute of Data Science, College Station, Texas, US
2022	*RevBayes Workshop*, Evolution conference, Cleveland, Ohio, US
2022	*Sequencing technician (RA)*, Agrilife Genomics & Bioinformatics Service, Texas A&M University, College Station, Texas, United States
2019	*Midwest Phylogenetics Workshop*, Itasca Field Station, Minnesota, United States

SKILLSETS

Coding Languages
Bash, Python, R

Bioinformatics
Bioinformatic pipeline development for processing short read whole genome sequences; parallel processing of largescale phylogenomic projects

Fieldwork
Ichthyological collection and preservation, botanical collection and preservation, tissue sampling and preservation for sequencing, elastomer tagging fishes for mark-recapture studies, field identifications of fishes, plants, and insects, DSLR camera use for morphometric photos, orienteering

Lab
Tissue preservation, sample preparation, DNA extractions (phenol chloroform and Qiagen), DNA quantification using Qubit and Nanodrop, PCR, primer design, library preparation

Spoken Languages
Fluent English, intermediate French, beginner Spanish

Citizenship
Canada, United States